

CW1 - Decision Tree

Jade Carpentier, Thomas Cherqui, Hannah O’Keeffe

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1 Visualization

Here is an example of output of the visualization function for a decision tree.

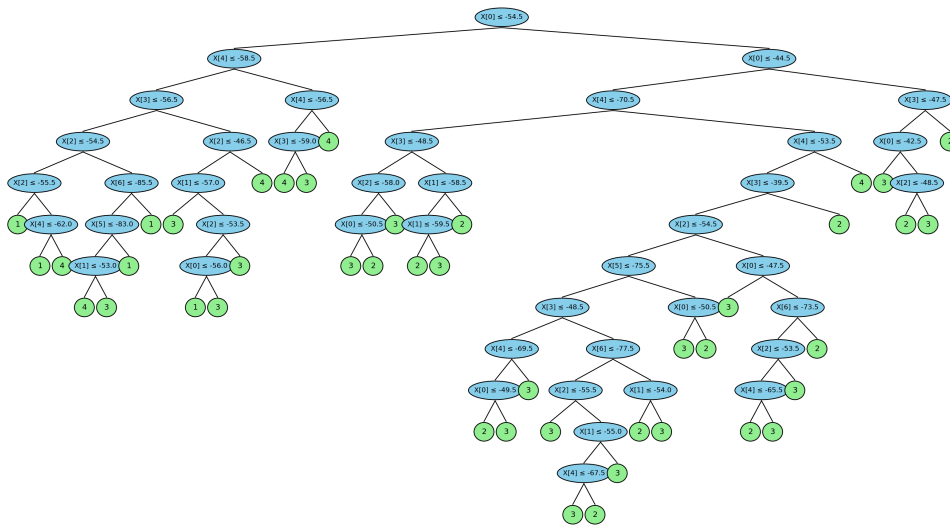


Figure 1: Decision Tree Visualisation Example

2 Evaluation

2.1 Cross validation classification metrics

By computing the average over all the test folds, we report the following cross-validation classification metrics for both clean and noisy data.



Figure 2: Global confusion matrices for the clean dataset (left) and the noisy dataset (right), where label corresponds to room number

	Clean Dataset				Noisy Dataset			
Class	Room 1	Room 2	Room 3	Room 4	Room 1	Room 2	Room 3	Room 4
Recall	0.983	0.966	0.960	0.990	0.797	0.791	0.811	0.794
Precision	0.989	0.963	0.958	0.988	0.805	0.797	0.780	0.812
F1	0.986	0.965	0.959	0.989	0.800	0.793	0.794	0.801

Table 1: Classification metrics for each class

Overall Accuracy:

Clean dataset: 97.5%

Noisy dataset: 79.8%

2.2 Result analysis

Comment for both datasets on which rooms are correctly recognised, and which rooms are confused with others.

In the clean dataset, the recall, precision and f1 scores are all high as most rooms are correctly predicted, with some confusion mainly between Room 2 and Room 3 (1.7 and 1.8), suggesting similar features between these classes. The noisy dataset shows increased misclassification across all rooms and in particular confusing Room 3 for Room 2.

2.3 Dataset differences

Is there any difference in the performance when using the clean and noisy datasets? If yes/no explain why

Accuracy drops from 97.5% to 79.8% when using the noisy dataset. This is likely due to incorrect labels, outliers and greater variation in feature values which introduce noise making it harder for the algorithm to generalise patterns in the data and classify correctly. The model is forced to make extra splits to fit noisy samples near value boundaries, which increases over fitting and reduces performance on the test data.

3 Pruning

3.1 Cross validation classification metrics



Figure 3: Global confusion matrices for the clean dataset (left) and the noisy dataset (right) with pruning

Overall Accuracy:

Clean dataset: 97.0%

Noisy dataset: 86.6%

	Clean Dataset				Noisy Dataset			
Class	Room 1	Room 2	Room 3	Room 4	Room 1	Room 2	Room 3	Room 4
Recall	0.992	0.946	0.946	0.986	0.875	0.867	0.843	0.880
Precision	0.979	0.960	0.939	0.993	0.868	0.871	0.855	0.878
F1	0.985	0.952	0.942	0.989	0.870	0.868	0.847	0.878

Table 2: Classification metrics for each class for pruned trees

3.2 Result analysis

Comment on the difference in performance before and after pruning for both datasets.

Pruning improves the noisy dataset accuracy from 79.8% to 86.6% and increases the recall, precision and f1 scores across all rooms by reducing over fitting as it decreases the tree’s depth and complexity. The clean dataset shows little change, as it contains fewer noisy examples meaning the unpruned tree already generalises well.

3.3 Depth Analysis

Comment on the average depth of the trees you generated for both datasets, before and after pruning. What can you tell about the relationship between maximal depth and prediction accuracy.

The tree depths averaged 13.6 (clean) and 19.8 (noisy), before pruning, and 7.9 (clean) and 12.3 (noisy), after. The reduced depths highlight how pruning removed branches capturing outliers in the data, allowing the trees to generalise better. There appears to be an optimal depth, after which accuracy plateaus or decreases due to over fitting.